



Lab 05

Series and Parallel RL & RC Circuits

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5.1 Objective

To examine the Impedance, voltage and current relationships in series and parallel RC and RL networks.

5.2 Pre-Lab - Task 1

1. Using Figure 5.1 with $V_{in} = 3\text{ V}$ sine at 10 kHz , $R = 1\text{ k}\Omega$, and $C = 10\text{ nF}$, determine the theoretical capacitive reactance (X_C) and circuit impedance ($Z \angle \theta$), and record the results in Table 5.2 and complete the calculated section of this table at other frequencies as well. Also, calculate the theoretical value of α and fill table 5.1.
2. Using Figure 5.2 with $V_{in} = 3\text{ V}$ sine at 10 kHz , $R = 1\text{ k}\Omega$, and $L = 10\text{ mH}$, determine the theoretical capacitive reactance (X_L) and circuit impedance ($Z \angle \theta$), and record the results in Table 5.4 and complete the calculated section of this table at other frequencies as well. Also, calculate the theoretical value of α and fill Table 5.4.
3. Using Figure 5.3 with $V_{in} = 3\text{ V}$ sine at 10 kHz , $R = 1\text{ k}\Omega$, and $C = 10\text{ nF}$, determine the theoretical capacitive reactance (X_C) and circuit impedance ($Z \angle \theta$), and record the results in Table 5.5 and complete the calculated section of this table at other frequencies as well.

Note: α is the theoretical angle between the V_C and I_C in table 5.1. Think about the relation between the voltage and current for capacitors! Similarly, apply this concept for β in table 5.3.

5.3 Introduction

Each element has a unique phase response: for resistors, the voltage is always in phase with the current, for capacitors the voltage always lags the current by 90 deg, and for inductors the voltage always leads the current by 90 deg. Consequently, a series combination of R, L, and C components will yield a complex impedance with a phase angle between +90 and -90 deg. Due to the phase response, KVL must be computed using vector (phasor) sums rather than simply relying on the magnitudes. Indeed, all computations of this nature, such as a voltage divider, must be computed using vectors.

5.4 Series RL and RC Circuits

Series RL (Resistor-Inductor) and RC (Resistor-Capacitor) circuits are two types of electrical circuits that consist of passive components, such as resistors, capacitors, and inductors, connected in series. These circuits have different characteristics and behaviors, particularly in response to AC (alternating current) signals. The inductor causes a phase shift of +90 degrees, while the capacitor causes a phase shift of -90 degrees.

5.5 Series RC Circuits

The behavior of a series RC circuit depends on the frequency of the applied AC signal:

DC (Direct Current): When a direct current is applied (frequency = 0), the capacitor acts as an open circuit (ideal capacitor) as it blocks the flow of direct current. Thus, the voltage drop across the capacitor is zero, and the entire voltage is dropped across the resistor.

AC (Alternating Current): As the frequency increases, the capacitive reactance ($X_c = \frac{-j}{2\pi fC}$) decreases, where f is the frequency and C is the capacitance of the capacitor. This results in decreasing opposition to the flow of current, leading to a phase shift between voltage and current. In an ideal capacitor, this phase shift is -90 degrees.

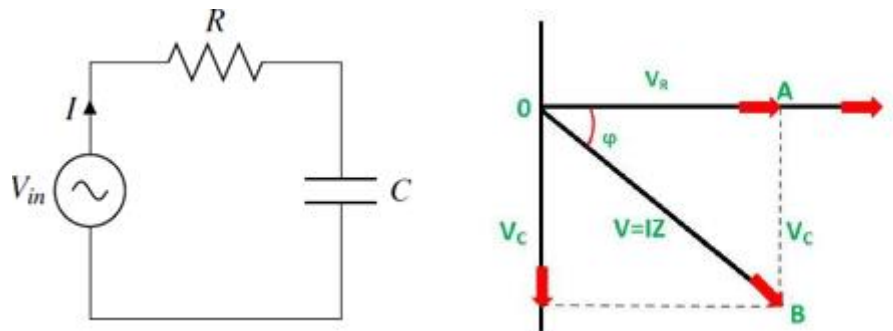


Figure 1: (a) Series RC Circuit, (b) Phasor Diagram

Formulae

$$X_c = \frac{-j}{2\pi fC} \quad (1)$$

$$|Z| = \sqrt{R^2 + X_c^2} \quad (2)$$

$$Z \angle \theta = \tan^{-1} \left(\frac{X_c}{R} \right) \quad (3)$$

$$V_c = \left(\frac{-jX_c}{-jX_c + R} \right) V_{in} \quad (4)$$

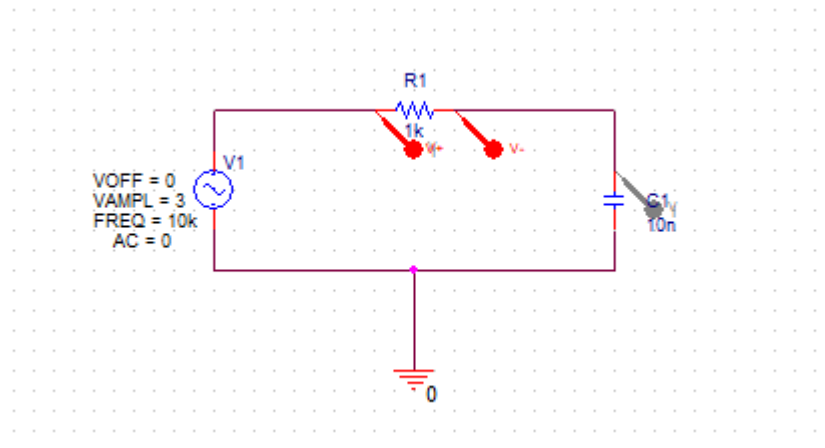
$$\alpha = \Delta t \times 360 \times freq \quad (5)$$

Task 2: To examine the voltage and current relationship of series RC circuits on PSpice

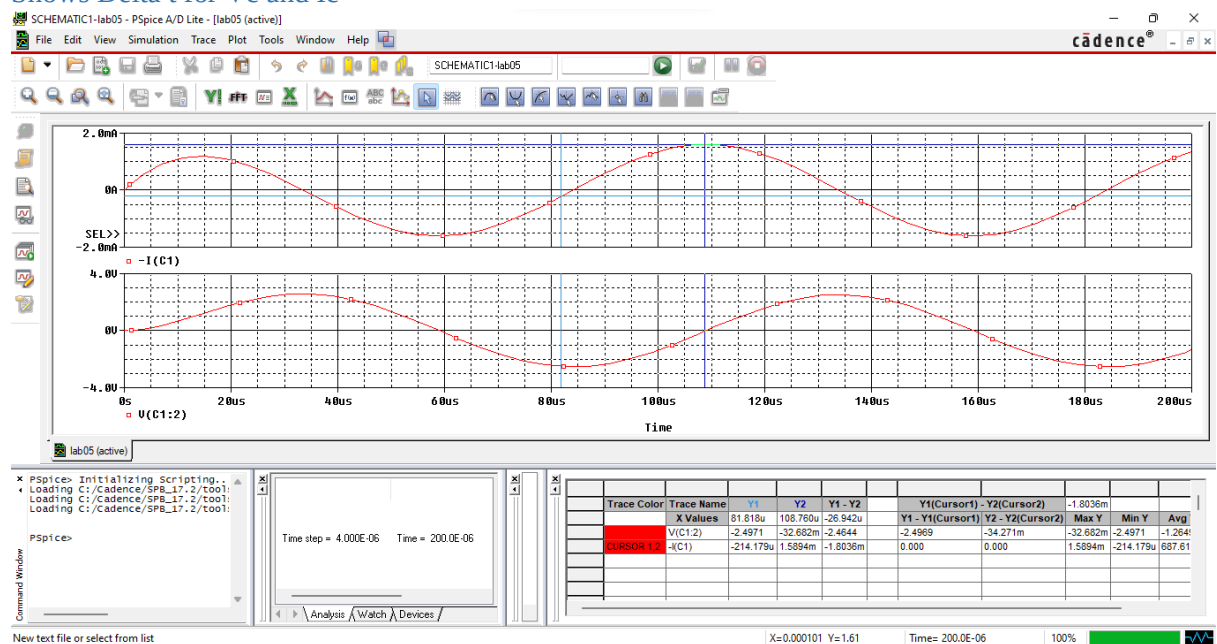
- Using Figure 5.1 with $V_{in} = 3V$ sine at $10kHz$, $R = 1k\Omega$, and $C = 10nF$, determine the theoretical capacitive reactance (X_c) and circuit impedance ($Z \angle \theta$), and record the results in Table 5.2.

- Construct the circuit of Figure 5.1 on PSpice. Use Sine Source (VSIN) as input with 3V amplitude [Keep AC = 0].
- Run Time-Domain (Transient) Analysis and obtain plots for Capacitor Voltage (V_C) and Resistor Voltage (V_R)¹ in the same plot window. Add another plot, obtain plot for current flowing through the circuit, use either $I(C1)$ or $I(R1)$.

Schematic

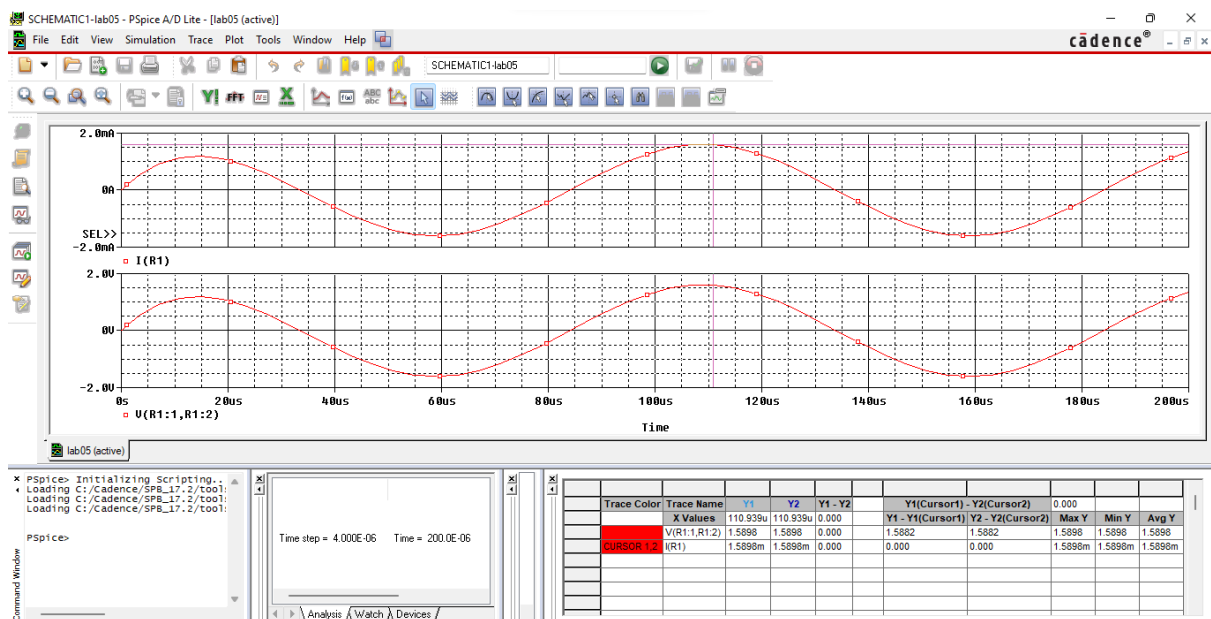


Shows Delta t for V_C and I_C



Shows Delta t for V_R and I_R which is 0.

¹ For the voltage across resistor use differential marker option in PSpice.



4. Determine phase difference between Voltage and Current for Capacitor and Resistor. Use markers in Current and Voltage plots to find Δt between V_c and I_c . Determine the value of α simulated using Δt . Record the results in Table 5.1. Similarly, repeat this for Resistor and record results in Table 5.1. (For Δt consider the starting point of each waveform).

$$\Delta t = 26.942\mu$$

$$\alpha = \Delta t \times 360 \times 10k$$

$$\alpha = 96.66^\circ$$

Table 5.1

*	For Resistor	For Capacitor
α Theoretical	0	90 degrees
α Simulated	0	96.66 degrees
Leading component [Voltage/Current]	-	Current

5. Analyze the results obtained in Table 5.1 below.

For Resistor there is no change in simulated and theoretical, However, there is a 6.6 degrees change for capacitor in simulated as compared to theoretical. It could be human error in reading the value of change in time but there could be additional voltages from the wires hindering the circuit.

6. For overall circuit phase theta θ , use markers in Current and Voltage plots to find Δt between V_{source} and I . Use appropriate markers to fill Table 5.2 for 10 kHz parameter. Find X_C Sim and



$Z\angle\theta$ Sim values by putting simulated values of V_C , I_T and V_{in} in Ohm's Law ($X_C = \frac{V_C}{I}$, $Z = \frac{V_{in}}{I}$) (Consider Peak Value for all the readings).

$V_{in}(\text{sim}) = 2.9642 \text{ V}$

- For angle $Z\angle\theta$ Sim, use formula i.e., $\theta = \Delta t \times 360 \times \text{freq}$. Using the total value of $Z\angle\theta$ Sim you may find the value of $I_T \angle\theta$ Sim.
- Repeat Transient analysis for remaining Table 5.2 frequency values.
- In the V_{in} source parameters, change the value $AC = 3$. Run AC Sweep for 10 Hz to 100 kHz. Use logarithmic scale. Attach the screenshots of Plots (I_T , phase of I_T , V_C , X_C , Z) w.r.t to f and provide your analysis on each plot.

Note: Think about why this reference for Δt is taken.

Table 5.2

Freq (Hz)	V_C sim	$I_T \angle\theta$ Sim	Δt	X_C Th.	X_C Sim.	$Z \angle\theta$ Th. (V_{in}/I)	$Z \angle\theta$ Sim.
10 k	2.5317	1.5898m<55.33	15.37u	1591.55	1592.46	1879.64 < 57.87	1887.03<-55.33
20 k	1.8680	2.3474m<36	219.334n	795.77	795.77	1277.98 < 38.5	1278<-36
50 k	909.807m	2.8873m<15.02	834.850n	318.33	315.10	1049.4 < 17.66	1039.03<-15.02
100 k	476.675m	2.9633m<8.064	224.026n	159.15	160.86	1012.6 < 9.04	1012.4<-8.064

As the frequency is increasing the Total Current is increasing which in turn decreases the reactive capacitance which means that circuit is becoming more resistive.

The phase shift is also decreasing which also shows the circuits approach to resistiveness.

As a result, overall impedance is decreasing.

5.6 Series RL Circuits

The behavior of a series RL circuit depends on the frequency of the applied AC signal:

DC (Direct Current): When a direct current is applied (frequency = 0), the inductor acts as a short circuit (ideal inductor) as the current takes some time to build up. Thus, the voltage drop across the inductor is zero, and the entire voltage is dropped across the resistor.

AC (Alternating Current): As the frequency increases, the inductive reactance ($X_L = j2\pi fL$) increases, where f is the frequency and L is the inductance of the coil. This results in an increasing opposition to the flow of current, leading to a phase shift between voltage and current. In an ideal inductor, this phase shift is 90 degrees.

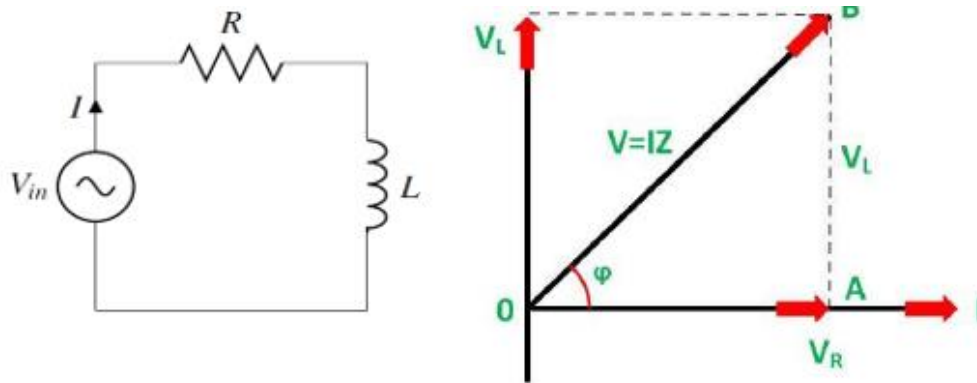


Figure 5.2: a) Series RL Circuit, (b) Phasor Diagram

Formulae

$$X_L = j2\pi fL \quad (6)$$

$$|Z| = \sqrt{R^2 + X_L^2} \quad (7)$$

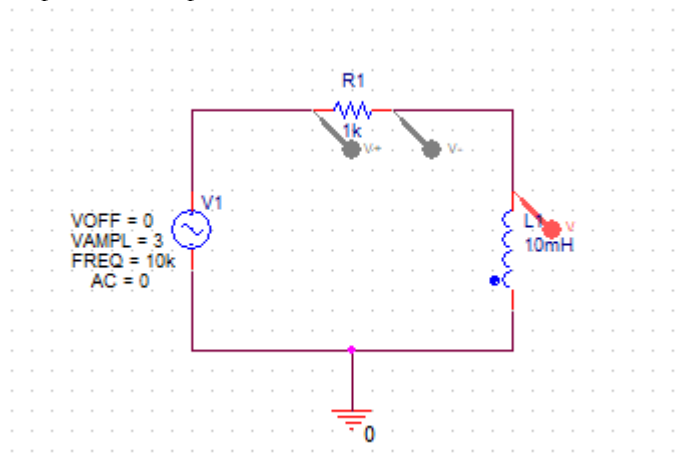
$$Z \angle \theta = \tan^{-1} \left(\frac{X_L}{R} \right) \quad (8)$$

$$V_L = \left(\frac{jX_L}{jX_L + R} \right) V_{in} \quad (9)$$

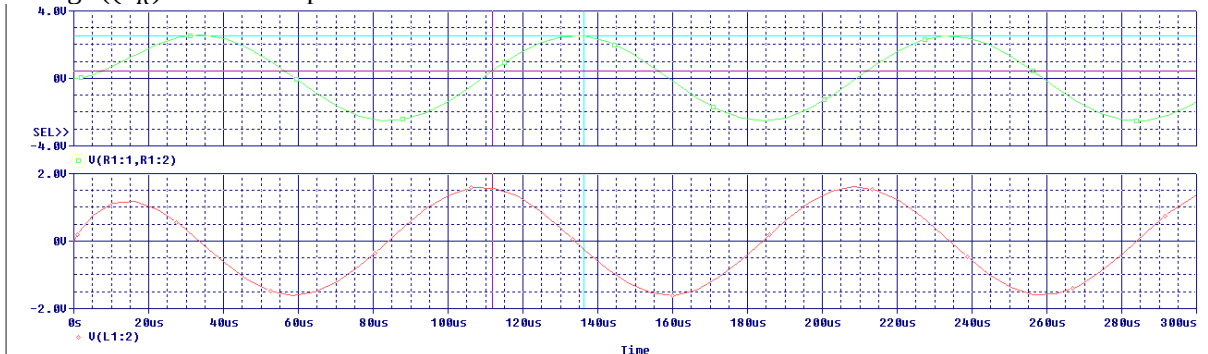
$$\beta = \Delta t \times 360 \times freq \quad (10)$$

Task 3: To examine the voltage and current relationship of series RL circuit

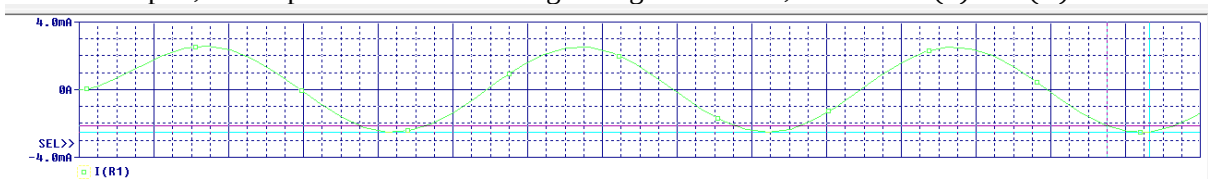
- Using Figure 5.2: a) Series RL Circuit, (b) Phasor DiagramFigure 5.2 with $V_{in} = 3\text{ V}$ sine at 10 kHz , $R = 1\text{ k}\Omega$, and $L = 10\text{ mH}$, determine the theoretical inductive reactance (X_L) and circuit impedance ($Z \angle \theta$), and record the results in Table 5.4.
- Construct the circuit of Figure 5.2 on PSpice. Use Sine Source (VSIN) as input with 3V amplitude [Keep AC = 0].



- Run Time-Domain (Transient) Analysis and obtain plots for Inductor Voltage (V_L) and Resistor Voltage (V_R) in the same plot window.



- Add another plot, obtain plot for current flowing through the circuit, use either $I(L)$ or $I(R)$.



- Determine phase difference between Voltage and Current for Inductor and Resistor. Use markers in Current and Voltage plots to find Δt between V_L and I_L . Determine the value of α simulated using Δt . Record the results in Table 5.3. Similarly, repeat this for Resistor and record results in Table 5.3. (For Δt consider the starting point of each waveform).

$$\Delta t = 24.444\mu$$

$$\alpha = \Delta t \times 360 \times 10k$$

$$\alpha = 87.99^\circ$$

Table 5.3

*	For Resistor	For Inductor
β Theoretical	0	90 degrees
β Simulated	0	87.99 degrees
Leading component [Voltage/Current]	-	Voltage

6. Analyze the results obtained in Table 5.3 below.

As seen there is no change in resistor however, there is a change of 2.01 degrees in simulated values and according to our simulation Voltage is leading.

7. For overall circuit phase theta θ , use markers in Current and Voltage plots to find Δt between V_{source} and I . Use appropriate markers to fill Table 5.4 for 10 kHz parameter. Find X_C Sim and $Z\angle\theta$ Sim values by putting simulated values of V_L , I_T and V_{in} in Ohm's Law ($X_L = \frac{V_L}{I}$, $Z = \frac{V_{in}}{I}$) (Consider Peak Value for all the readings). For angle $Z\angle\theta$ Sim, use formula i.e., $\theta = \Delta t \times 360 \times freq$. Using the total value of $Z\angle\theta$ Sim you may find the value of $I_T \angle\theta$ Sim.

Table 5.4

Freq (Hz)	V_L Sim	$I_T \angle\theta$ Sim	Δt	X_L Th.	X_L Sim.	$Z\angle\theta$ Th.	$Z\angle\theta$ Sim.
10 k	1.5898	2.4506m<40.85	11.346u	628.32	648.74	1181.03 < 32.14	1224.19<-40.85
20 k	2.3651	1.8344m<45.48	6.3162u	1256.63	1289.30	1605.96 < 51.48	1635.41<-45.48
50 k	2.8377	905.828u<76.65	4.2584u	3141.6	3132.71	3296.9 < 72.34	3311.88<-76.65
100 k	2.8171	461.410u<78.84	2.1901u	6283.2	6105.42	6362.3 < 80.96	6501.8<-78.84

8. Repeat Transient analysis for remaining Table 5.4 frequency values.
9. In the V_{in} source parameters, change the value $AC = 3$. Run AC Sweep for 10 Hz to 100 kHz. Use logarithmic scale. Attach the screenshots of Plots (I_T , phase of I_T , V_L , X_L , Z) w.r.t to f .
10. Discuss your analysis for each plot below in detail.

As the frequency is increasing the Total Current is decreasing which in turn increases the reactive inductance which means that circuit is becoming more resistive.

The phase shift is also increasing which also shows the circuits approach to resistiveness. 

As a result, overall impedance is increasing.

Parallel RL and RC Circuits

5.5 Parallel RC Circuit

In a parallel RC circuit, the line current leads the applied voltage by some phase angle less than 90 degrees but greater than 0 degrees. The exact angle depends on whether the capacitive current or resistive current is greater. If there is more capacitive current, the angle will be closer to 90 degrees, while if the resistive current is greater, the angle is closer to 0 degrees.

The value of the phase angle can be calculated from the values of the two branch currents using equation (14):

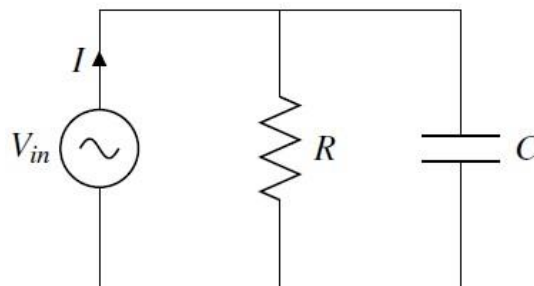


Figure 5.3: Parallel RC Circuit

Formulae

$$X_C = \frac{-j}{2\pi fC} \quad (11)$$

$$I_T = \sqrt{I_R^2 + I_C^2} \quad (12)$$

$$Z = \frac{R X_C^2}{R^2 + X_C^2} - \frac{jR^2 X_C}{R^2 + X_C^2} \quad (13)$$

$$\phi = \tan^{-1}\left(\frac{I_C}{I_R}\right) \quad (14)$$

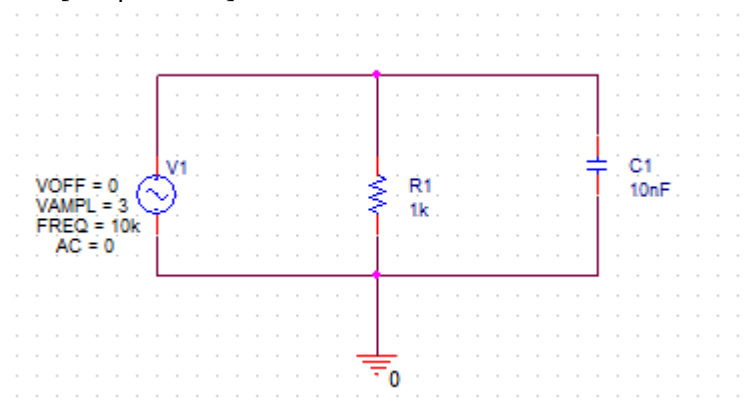
$$I_C = \frac{V}{-jX_C} \quad (15)$$

$$Z = \frac{V}{I_T} \quad (16)$$

$$\phi = \Delta t \times 360^\circ \times f \quad (17)$$

Task 4: To examine the voltage and current relationship of parallel RC circuits

- Using Figure 5.2 with $V_{in} = 3V$ sine at 10 kHz , $R = 1\text{ k}\Omega$, and $C = 10\text{ nF}$, determine the theoretical capacitive reactance (X_C) and circuit impedance ($Z \angle \theta$), and record the results in Table 5.5.
- Construct the circuit of Figure 5.3 on PSpice. Use Sine Source (VSIN) as input with 3V amplitude [Keep AC = 0].

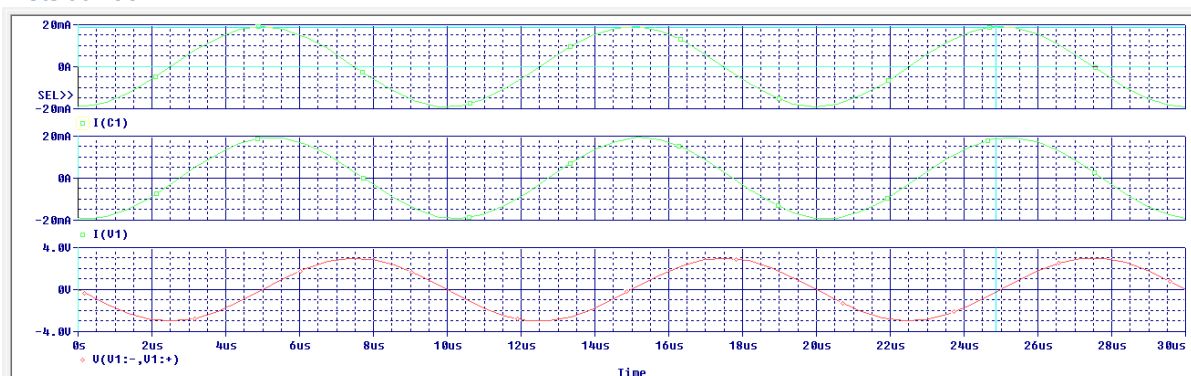


- Run Time-Domain (Transient) Analysis, obtain plots for Source Voltage (V_{in}) in one plot

window and Total current (I_T), and capacitor current (I_C) in different plot window. Record the peak value of total current and Capacitor Current in Table 5.5.

4. Determine phase difference between Input Source Voltage and Total Current. Use markers in Current and Voltage plots to find Δt between V and I, using Δt determine the value of θ simulated and record in Table 5.5

Plots at 100kHz



5. Based on the simulated values, determine the simulated Z and X_C values using Ohm's Law ($X_C = \frac{V_C}{I}$, $Z = \frac{V_{in}}{I}$) and record in Table 5.5.
6. Repeat Transient analysis for remaining frequency values.
7. (Optional) In the Vin source parameters, change the value AC = 3. Run AC Sweep for 5 Hz to 100 kHz. Use logarithmic scale. Attach the screenshots of Plots (I_T , phase of I_T , V_C , X_C , Z) w.r.t to f .

Table 5.5

Freq.	$I_T \angle \theta$ (Sim.)	Δt	I_C (Sim.)	X_C (Sim.)	$Z \angle \theta$ (Th.)	$Z \angle \theta$ (Sim.)
5K	3.1341m<90	50.000u	625.017n	4.799u	910.16-j285.9	957.2<-90
20K	4.7766m<24.62	6.8385u	3.8072m	787.98	387.72-j487.23	628.06<-24.6
50K	9.943m<70.0	3.8934u	9.4419m	317.73	91.99 - j289.02	301.72<-70
100K	19.154m<87.58	2.4327u	18.866m	159.02	24.7 - j155.22	156.63<-87.5

8. Discuss your analysis for each plot/table 5.5 below in detail.

As the frequency increases Total Current increases, and as we can see there is an increase in Capacitor current. We can also observe a decrease in Reactive Capacitance of the overall circuit and due to which the Impedance decreases. The value of phase shift is getting closer and closer to 90 making the circuit resistive once it reaches 90 degrees.

5.7 Parallel RL Circuit

In parallel DC circuits, the simple arithmetic sum of the individual branch currents equals the total current. The same is true in an AC parallel circuit if only pure resistors or only pure inductors are connected in parallel. However, when a resistor and inductor are connected in parallel, the two currents will be out of phase with each other. In this case, the total current is equal to the vector sum rather than the arithmetic sum of the currents.

Recall that the voltage and current through a resistor are in phase, but through a pure inductor the current lags the voltage by exactly 90 degrees. This is still the case when the two are connected in parallel.

In all parallel RL circuits, the phase angle theta (θ) by which the total current lags the voltage is somewhere between 0 and 90 degrees. The size of the angle is determined by whether there is more inductive current or resistive current.

If there is more inductive current, the phase angle will be closer to 90 degrees. It will be closer to 0 degrees if there is more resistive current. From the circuit vector diagram, you can see that the value of the phase angle can be calculated from the equation:

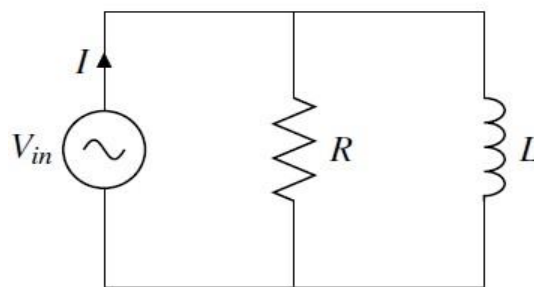


Figure 2: Parallel RL Circuit

Formulae

$$X_L = j2\pi fL \quad (18)$$

$$I_T = \sqrt{I_R^2 + I_L^2} \quad (19)$$

$$Z = \frac{R X_L^2}{R^2 + X_L^2} - \frac{jR^2 X_L}{R^2 + X_L^2} \quad (20)$$

$$\phi = \tan^{-1}\left(\frac{I_L}{I_R}\right) \quad (21)$$

$$I_L = \frac{V}{jX_L} \quad (22)$$

$$I_R = \frac{V}{R} \quad (23)$$

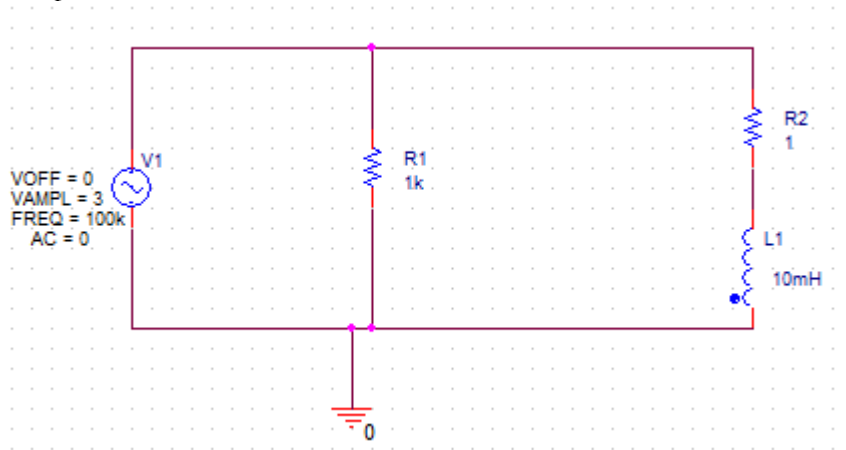
$$Z = \frac{V}{I_T} \quad (24)$$

$$\phi = \Delta t \times 360^\circ \times f \quad (25)$$

Post-Lab

Task 5: To examine the voltage and current relationship of parallel RL circuits

- Using circuit in Figure 5.4 with $V_{in} = 3V$ sine at 10 kHz , $R = 1\text{ k}\Omega$, and $L = 10\text{ mH}$, determine the theoretical capacitive reactance (X_L) and circuit impedance ($Z \angle \theta$), and record the results in Table 5.6.
- Construct the circuit of Figure 5.4, on PSpice. Use Sine Source (VSIN) as input with 3V amplitude [Keep AC = 0].



- Run Time-Domain (Transient) Analysis, obtain plots for Source Voltage (V_{in}) in one plot window and Total current (I_T), and Inductor current (I_L) in different plot window. Record the peak value of total current and Inductor Current in Table 5.6
- Determine phase difference between Input Source Voltage and Total Current. Use markers in Current and Voltage plots to find Δt between v and I , using Δt determine the value of θ simulated and record in Table 5.6
- Based on the simulated values, determine the simulated Z and X_L values using Ohm's Law ($X_L = \frac{V_L}{I}$, $Z = \frac{V_{in}}{I}$) and record in Table 5.6
- Repeat Transient analysis for remaining frequency values.
- (Optional)** In the V_{in} source parameters, change the value AC = 3. Run AC Sweep for 5 Hz to 100 kHz. Use logarithmic scale. Attach the screenshots of Plots (I_T , phase of I_T , V_C , X_C , Z) w.r.t to f .



Table 5.6

Freq.	$I_T \angle \theta$ (Sim)	Δt	I_L (Sim)	X_L (Sim)	$Z \angle \theta$ (Th.)	$Z \angle \theta$ (Sim).
5K	829.884u	62.307u	17.956m	167.08		3614.96<67.84
20K	1.410m	31.144u	4.7285m	634.45		2127.66<
50K	2.1507m	11.143u	1.8807m	1595.15		1394.89<19.6
100K	2.4501m	5.1551u	366.380u	8188.22		1224.44<5.58

8. Discuss your analysis for each plot/ Table 5.6 below in detail.

As the frequency increases Total Current increases, and as we can see there is an decrease in Inductor current. We can also observe a increase in Reactive Inductance of the overall circuit and due to which the Impedance decreases. The value of phase shift is getting closer and closer to 0 making the circuit resistive once it reaches 0 degrees.



Assessment Rubric Lab 05

Name:	Student ID:
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Points Distribution

Task No.	LR 2 Simulation	LR5 Results/Plots	LR6 Calculations	LR 10 Analysis
Task 1			/15	
Task 1	/5	/12.5		/5
Task 2	/5	/12.5		/5
Task 3	/2.5	/5		/2.5
Task 4	/2.5	/5	/5	/2.5
Total	/20	/40	/15	/10
CLO Mapped	CLO 1	CLO 1	CLO1	CLO 1

Affective Domain Rubric		Points	CLO Mapped
AR 7	Report Submission (Pre & Post Lab)	/(5+5)	CLO 4

CLO	Total Points	Points Obtained
1	90	
4	10	
Total	100	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.



Lab Evaluation Assessment Rubric

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3: Good Points 3	Level 4: Exemplary Points 4
LR1	Circuit Layout	Connections between circuit components are mostly wrong. Circuit layout is cluttered. Needs guidance to make correct equipment/ component connections.	Few of the connections made between circuit components are wrong. Circuit layout is not neat and clean as per standards.	Circuit layout and connections are correct but not all connections are neat and clean as per standards.	Neat, clean and correct connections of all the circuit components/ equipment are made as per standard circuit diagram.
LR2	Program/Code/ Simulation Model/ Network Model	Program/code/simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR3	Troubleshooting	Unable to identify the fault/minimal effort show in troubleshooting.	Able to identify the fault but unable to remove it.	Able to identify the fault but partially removes it.	Able to identify the fault and takes necessary steps and actions to correct it.
LR4	Data Collection	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included.	Measurements are somewhat inaccurate and imprecise. Observations are incomplete or vague. Major errors are there in using symbols, units and significant digits.	Measurements are mostly accurate. Observations are generally complete. Minor errors are present in using symbols, units and significant digits.	Measurements are both accurate and precise. Data collection is systematic. Observations are very thorough and include appropriate symbols, units and significant digits and task completed in due time.
LR5	Results & Plots	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR6	Calculations	Formulae used and/or calculations are incorrect. Units are not mentioned with results.	Formulae used are correct but there are few mistakes in calculation which lead to incorrect results.	Formulae used and end results are correct. Complete working steps are not shown. Proper units are not mentioned with results.	Formulae used, end results and all calculations are correct with all the intermediate steps clearly shown. Units are mentioned with results.
LR9	Report	All the in-lab tasks are not included in report and / or the report is submitted too late.	Most of the tasks are included in report but are not well explained. All the necessary figures / plots are not included. Report is submitted after due date.	Good summary of most the in-lab tasks is included in report. The work is supported by figures and plots with explanations. The report is submitted timely.	Detailed summary of the in-lab tasks is provided. All tasks are included and explained well. Data is presented clearly including all the necessary figures, plots and tables.
LR10	Analysis	Results are not interpreted or are analyzed incorrectly.	Analysis of the obtained results is incomplete. Conclusions are not drawn.	Results are analyzed and concluded but are not well explained and justification is not provided with the help of reasoning.	Results are well analyzed supported with reasons. Good comparison is made between the theoretical and experimental results with correct and useful conclusion.
LR11	Design	Proposed design is unsubstantiated or does not satisfy most of given constraints. The breakdown of system into features is incomplete and still at lower resolution.	Justification for proposed design is weak, and it only satisfies some constraints. The breakdown of system is missing some features and resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies most given constraints. The breakdown of system into features seems complete, but resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies all given constraints. The breakdown of system into features seems complete and at appropriate resolution, given students' level.